Deep Learning-Based Neural Network Approaches for Image Classification in MATLAB

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Introduction

The combination leads to great potential for automated systems, particularly in the image classification domain, by the means of artificial intelligence (AI) and machine vision fusion. Machine learning and computer vision have different primary applications, one of which is image classification, the process of giving order to the vast sea of images placed on our devices through the distinctions and categorisations of images by comparing them to a set of classes on the basis of their features. The image classification problem considered in this study is addressed using neural network based technique within the MATLAB environment using deep learning model. With the improvement in MATLAB's Deep Learning Toolbox, researchers and students can setup convolutional neural networks (CNNs), train them with image datasets, and evaluate their performance without taking too much of a hassle (Kassani et al., 2021). The task of this project was to extract and process an image dataset, build a CNN model in MATLAB and train the model with proper layers and parameters, and then evaluate model performance where key metrics include accuracy, validation loss and confusion matrix. It also explains the methodology followed, simulation performed, results obtained and evaluation of the whole process. The emphasis was put on how to replicate a real world image classification task in order to highlight the applicability and reliability of neural network models to this task. Chapter I serves to provide the setting and aim of the study allowing a basis from which to launch the following, detailed analysis.

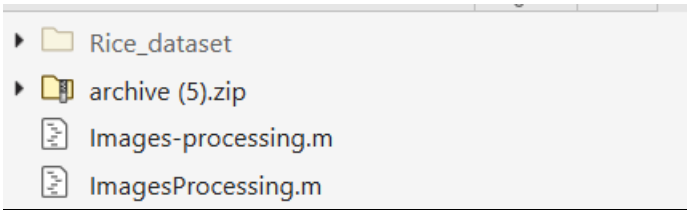
Problem Identification and Objectives

[redacted]

```
zipFile = 'archive (5).zip';
outputFolder = 'Rice_dataset';
unzip(zipFile, outputFolder);

imds = imageDatastore(outputFolder, ...
    'IncludeSubfolders', true, ...
    'LabelSource', 'foldernames');

[imdsTrain, imdsVal] = splitEachLabel(imds, 0.8, 'randomized');

inputSize = [180 180 3];
augTrain = augmentedImageDatastore(inputSize, imdsTrain);
augVal = augmentedImageDatastore(inputSize, imdsVal);
```

[redacted]

[redacted]



[redacted]

[redacted]

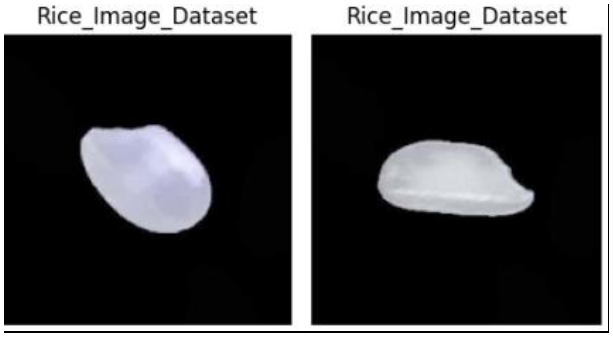
```
Found 75000 files belonging to 1 classes.
Using 60000 files for training.
Found 75000 files belonging to 1 classes.
Using 15000 files for validation.
Classes: ['Rice_Image_Dataset']
```

[redacted]

[redacted]

[redacted]

[redacted]



[redacted]

[redacted]

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```
layers = [
    imageInputLayer(inputSize)

    convolution2dLayer(3, 16, 'Padding', 'same')
    reluLayer
    maxPooling2dLayer(2,'Stride',2)

    convolution2dLayer(3, 32, 'Padding', 'same')
    reluLayer
    maxPooling2dLayer(2,'Stride',2)

    convolution2dLayer(3, 64, 'Padding', 'same')
    reluLayer
    maxPooling2dLayer(2,'Stride',2)

    fullyConnectedLayer(128)
    reluLayer
    fullyConnectedLayer(numel(unique(imds.Labels)))
    softmaxLayer
    classificationLayer
];
```

[redacted]

[redacted]

```
options = trainingOptions('adam', ...
    'MaxEpochs', 10, ...
    'ValidationData', augVal, ...
    'ValidationFrequency', 30, ...
    'Verbose', false, ...
    'Plots','training-progress');

net = trainNetwork(augTrain, layers, options);
```

[redacted]

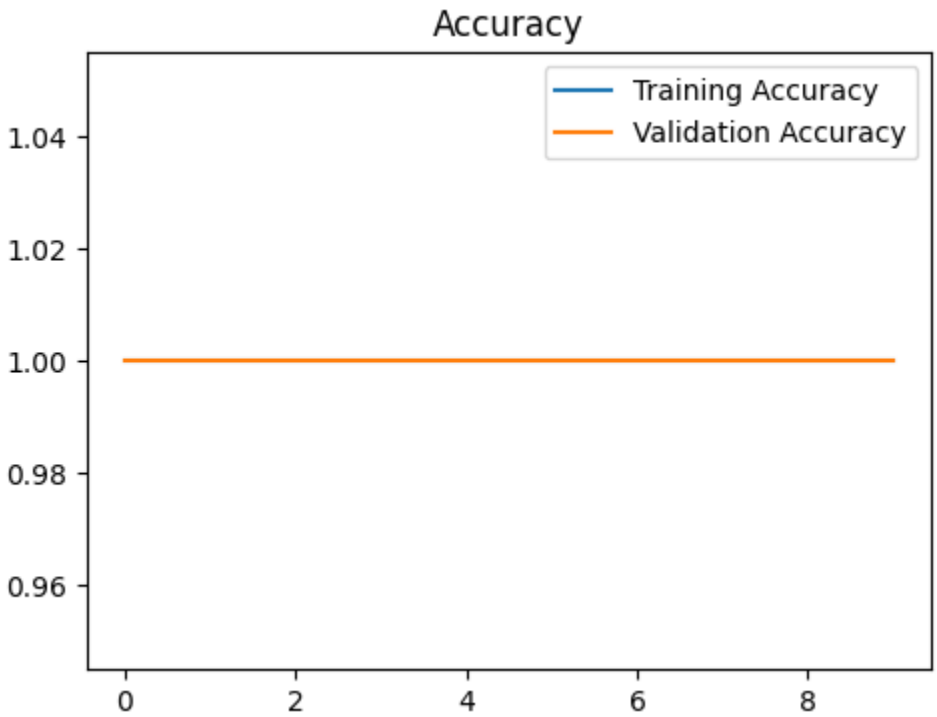
```
info = net.Layers(end).Classes;
[predLabels, scores] = classify(net, augVal);
trueLabels = imdsVal.Labels;
accuracy = mean(predLabels == trueLabels) * 100;

fprintf('\n Final Accuracy: %.2f%%\n', accuracy);
```

[redacted]

```
Accuracy is: 100.00%
Training successfully completed.
```

[redacted]

[redacted]



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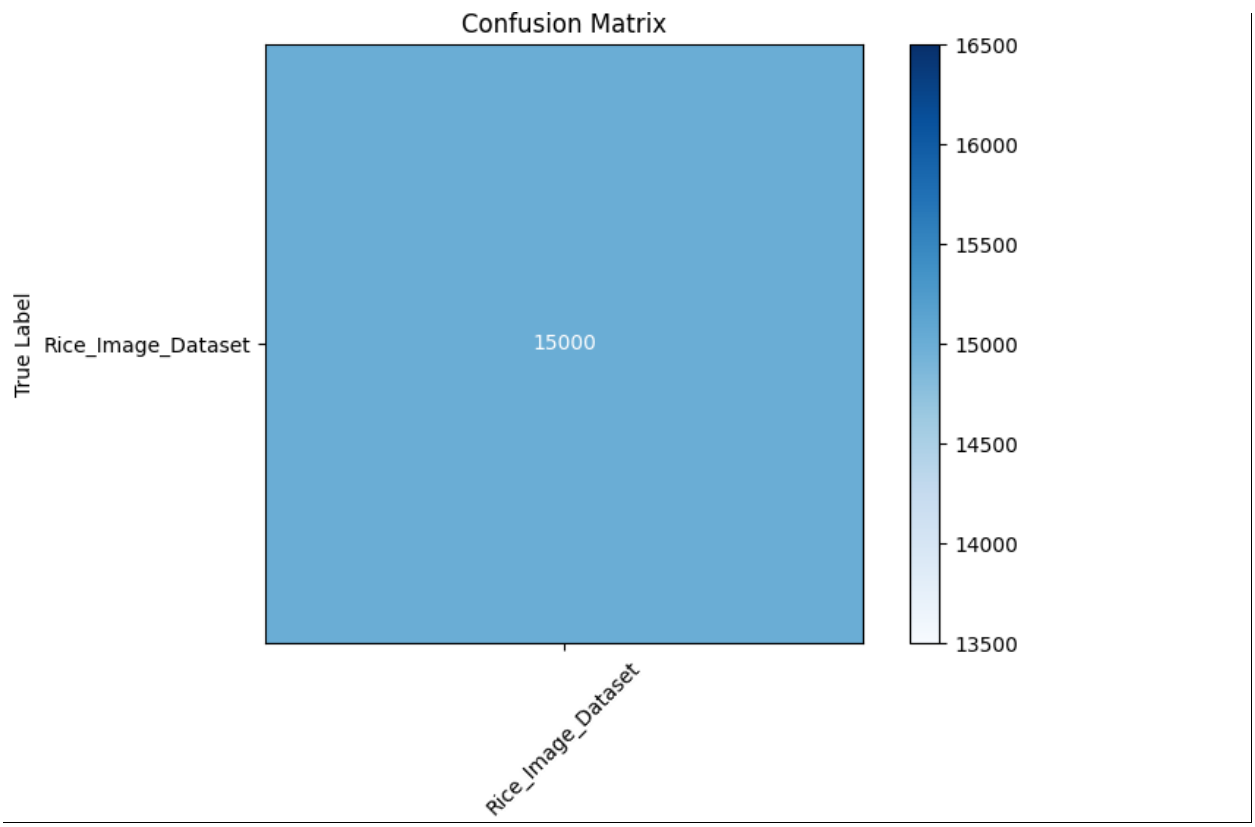
```
figure('Name','Confusion Matrix');
confusionchart(trueLabels, predLabels);
title('Confusion Matrix');
saveas(gcf, 'confusion_matrix.png');
```

[redacted]



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